

THE KLEIN QUARTIC

Most of this is based off of Noam Elkies' article on the Klein quartic, "The Klein Quartic in Number Theory".

1. THE KLEIN QUARTIC AS A RIEMANN SURFACE

The Klein quartic X is a projective curve in $\mathbf{P}^2(\mathbf{C})$ cut out by

$$x^3y + y^3z + z^3x = 0.$$

This is a compact Riemann surface of genus three, and in fact has 168 automorphisms.

Let's quickly think about these automorphisms. First, there are two obvious types of automorphisms:

- We can cyclically permute the variables, so there is a copy of $\mathbf{Z}/3\mathbf{Z}$ in $\text{Aut}(X)$ generated by an element A .
- A bit less obvious: fix a primitive 7th root of unity ζ . Then applying

$$B = \begin{pmatrix} \zeta^4 & & \\ & \zeta^2 & \\ & & \zeta \end{pmatrix}$$

to the vector (x, y, z) we obtain an order seven automorphism.

These first two generators satisfy $B^4 = ABA^{-1}$. Thus they generate a semidirect product of $\mathbf{Z}/3\mathbf{Z}$ and $\mathbf{Z}/7\mathbf{Z}$ of order 21.

There is also a highly non-obvious involution on X , given by applying

$$C = -\frac{1}{\sqrt{-7}} \begin{pmatrix} \zeta - \zeta^6 & \zeta^2 - \zeta^5 & \zeta^4 - \zeta^3 \\ \zeta^2 - \zeta^5 & \zeta^4 - \zeta^3 & \zeta - \zeta^6 \\ \zeta^4 - \zeta^3 & \zeta - \zeta^6 & \zeta^2 - \zeta^5 \end{pmatrix}.$$

Modulo the scaling factor, this is the Fourier transform on the space of odd functions $\mathbf{F}_7 \rightarrow \mathbf{C}$.

The group of automorphisms generated by $\langle A, B, C \rangle$ is now actually quite large, as all the 49 elements

$$B^a C B^b$$

for $a, b \in [0, 6]$ are in fact distinct automorphisms. In fact, this gives an explicit presentation of the unique simple group of order 168, $\text{PSL}_2(\mathbf{F}_7)$.

This is the largest possible number of automorphisms for this genus, meeting the Hurwitz bound.

PROPOSITION 1.1. Let X be a compact Riemann surface of genus $g \geq 2$. Then the group $\text{Aut}(X)$ of orientation-preserving conformal automorphisms has order at most

$$|\text{Aut}(X)| \leq 84(g - 1).$$

Sketch. Take X and set $G = \text{Aut}(X)$. The orbit space X/G has an induced complex structure from X , which in fact makes X/G a Riemann surface. The map

$$X \rightarrow X/G$$

is a branched covering, with finitely many ramification points (say k of them).

Let g_0 be the genus of X/G . Riemann-Hurwitz tells us

$$2g - 2 = |G| \cdot \left(2g_0 - 2 + \sum_{i \leq k} 1 - \frac{1}{e_i} \right).$$

The ramification indices at a ramification point are the orders of the stabilizers of that orbit. The number of preimages f_i of a ramification point then satisfies $e_i f_i = |G|$ by orbit-stabilizer.

We now need some casework. If $g_0 \geq 2$, then we get $2g - 2 \geq 2|G|$ so $|G| \leq g - 1$ and cannot be very large. In particular, the upper bound will want $g_0 = 0$. Also, if we have many ramification points (say ≥ 5) even if $g_0 = 0$ then we again get a good bound $|G| \leq 4(g - 1)$.

One can argue the least number of ramification points is 3 (χ cannot be negative, so if $k \leq 2$ there is no hope of making $-2 + \sum_{i \leq k} 1 - \frac{1}{e_i}$ positive), and then minimizing $1 - \frac{1}{e_1} - \frac{1}{e_2} - \frac{1}{e_3}$ among positive values we see the minimum is at $(2, 3, 7)$. The corresponding bound is

$$2g - 2 \geq |G| \cdot \left(\frac{1}{42} \right)$$

so $84(g - 1) \geq |G|$. □

A conceptual argument can also be given as follows. By uniformization, we can see X is covered by $\mathbb{H}$. The conformal maps on X are induced by orientation-preserving automorphisms of $\mathbb{H}$, so we want to maximize these. By Gauss-Bonnet, we see from the fact that there is no boundary and that the surface is hyperbolic that

$$\text{Area}(X) = -2\pi\chi(X) = 4\pi(g - 1).$$

We imagine X as coming from folding up a subset of $\mathbb{H}$ using the covering map, and that this subset is tiled from a fundamental domain D by applying automorphisms on $\mathbb{H}$. To

get the most automorphisms, we want D to be as small as possible. If D is a triangle with angles π/e_i , then we wish to minimize

$$\text{Area}(D) = \pi \left(1 - \sum \frac{1}{e_i} \right).$$

This is achieved for $(2, 3, 7)$, giving

$$\text{Area}(X)/\text{Area}(D) \leq 168(g-1).$$

This overcounts by a factor of two because the triangle domain is a fundamental domain for the full triangle group with reflections; the orientation-preserving subgroup has twice the area. If we account for this, we get the actual bound.

In fact, this fact about maximality of automorphisms uniquely characterizes the Klein quartic.

THEOREM 1.2. There is a unique genus 3 Riemann surface X with 168 automorphisms. We can equivalently characterize such an X by saying that the quotient map by its automorphism group

$$X \rightarrow X/\text{Aut}(X) \simeq \mathbf{P}^1$$

is a Galois branched cover ramified at three points with indices 2, 3, and 7.

2. AS A MODULAR CURVE

The previous criterion gives us an easy way to check if a curve is the Klein quartic over $\mathbf{C}$. However, in defining X we actually did so over $\mathbf{Q}$.

Modular curves are constructed as quotients of the upper half plane $\mathbb{H}$ by congruence subgroups. Namely, taking the usual action of linear fractional transformations

$$z \mapsto \frac{az + b}{cz + d}$$

we obtain an action of $\text{SL}_2(\mathbf{Z})$ on $\mathbb{H}$. A subgroup $\Gamma \leq \text{SL}_2(\mathbf{Z})$ is called a *congruence subgroup* if it contains $\Gamma(N)$ for some N , the subgroup

$$\Gamma(N) := \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbf{Z}) : a \equiv d \equiv 1 \pmod{N}, b, c \equiv 0 \pmod{N} \right\}.$$

Or, more simply, the kernel of the surjective map

$$\text{SL}_2(\mathbf{Z}) \rightarrow \text{SL}_2(\mathbf{Z}/N\mathbf{Z}).$$

We can use the chinese remainder theorem to see the map is surjective, so $\text{SL}_2(\mathbf{Z})/\Gamma(N) \simeq \text{SL}_2(\mathbf{Z}/N\mathbf{Z})$ which easily lets us compute the index.

DEFINITION 2.1. The affine modular curve $Y(N)$ is the quotient $\mathbb{H}/\Gamma(N)$. The compactified modular curve is

$$X(N) := \mathbb{H}^*/\Gamma(N)$$

where we add the cusps $\mathbf{Q} \cup \infty = \mathbf{P}^1(\mathbf{Q})$ to $\mathbb{H}$.

EXAMPLE 2.2. The curve $X(1)$ is $\mathbf{P}^1(\mathbf{C})$.

Observe that $\text{Aut}(X)$ is a simple group of order 168, so in particular it is isomorphic to $\text{PSL}_2(\mathbf{F}_7)$. The previous discussion shows that

$$\Gamma(1)/\Gamma(7) \simeq \text{SL}_2(\mathbf{F}_7).$$

Observing the matrices $\pm I$ induce the same automorphism, this shows that over $\mathbf{C}$ the modular curve $X(7)$ has an action by $G = \text{PSL}_2(\mathbf{F}_7)$.

COROLLARY 2.3. Over $\mathbf{C}$, the Klein quartic is isomorphic to $X(7)$ as a Riemann surface.

Proof. We actually have several ways to do this at this point. It is easy to check that $X(7)$ has at least 168 distinct automorphism, so by our previous results we just need to show it has genus 3.

The covering

$$X(7) \rightarrow X(1) \simeq \mathbf{P}^1$$

is Galois, with Galois group $G = \text{PSL}_2(\mathbf{F}_7)$. Applying Riemann-Hurwitz,

$$\chi(X(7)) = 168\chi(\mathbf{P}^1) - \sum_p (e_p - 1)$$

where the sum is over the ramification points $p \in X$. Since the cover is Galois, the ramification indices of preimages of a point in $\mathbf{P}^1$ are the same. Using the standard description of a fundamental domain for $X(1)$, the potential ramification points are $[i]$, $[\rho]$ (ρ is $e^{2\pi i/3}$) and $[\infty]$. The ramification indices here are 2, 3, and 7 respectively, computed by the size of stabilizer of the G -action at that point.

Note that just the ramification computation is enough to see we get X . □

In fact, this isomorphism even descends over $\mathbf{Q}$ when we use the $\mathbf{Q}$ -scheme $x^3y + y^3z + z^3x = 0$ for X .

THEOREM 2.4. Over $\mathbf{Q}(\zeta_7)$, the modular curve $X(7)$ parameterizes elliptic curves with full level 7 structure, i.e. a basis of $E[7]$ compatible with the Weil pairing. Over $\mathbf{Q}$, the relevant model is obtained by Galois descent from this full level structure.

REMARK 2.5. This is a special feature over $\mathbf{Q}$ that we need to think about the Galois action. The Weil pairing identifies $\Lambda^2 E[N] \simeq \mu_N$, so the descent data for a full level structure is part of the choice of the $\mathbf{Q}$ -model.

THEOREM 2.6. Over $\mathbf{Q}$, $X(7)$ is isomorphic to $x^3y + y^3z + z^3x = 0$.

The basic idea is that we can find some explicit weight 2 cusp forms for $\Gamma(7)$, or functions f such that

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^2 f(\tau)$$

for matrices in $\Gamma(7)$, and such that f vanishes at the cusps. Equivalently, $f(\tau)d\tau$ is $\Gamma(7)$ -invariant on the open modular curve, and the cuspidality condition is exactly what makes it extend across the compactification. Thus these cusp forms correspond to sections in $H^0(X(7), \Omega^1)$ (or differential forms). In particular, we find three such global sections which generate Ω^1 , which then defines a map

$$X(7) \rightarrow \mathbf{P}^2.$$

The cusp forms $x(\tau), y(\tau)$ and $z(\tau)$ satisfy exactly the relation of the Klein quartic, and moreover we can check this is an embedding. It turns out this strategy also descends to $\mathbf{Q}$.

Explicitly, these cusp forms take the form

$$\sum_{\beta \in \mathbf{Z}[\frac{-1+\sqrt{-7}}{2}]} \operatorname{Re}(\beta) q^{\beta\bar{\beta}/7}$$

where $q = e^{2\pi i\tau}$ and the sum runs over β congruent modulo $\sqrt{-7}$ to one of 1, 2, 4.

REMARK 2.7. To really descend to $\mathbf{Q}$ we need to check the moduli problem matches. To do this one can write down a generic elliptic curve attached to a non-cusp point $(x : y : z)$ on X , and then compute the Galois action on $E[7]$ to verify that the descended level structure is the desired one.

3. HEEGNER NUMBERS

The Stark-Heegner theorem is the following result.

THEOREM 3.1. The only imaginary quadratic fields with absolute discriminant D whose rings of integers are PIDs occur when

$$D = 3, 4, 7, 8, 11, 19, 43, 67, 163.$$

The basic idea of the argument is that the existence of such D implies the existence of a special elliptic curve. This follows from the theory of complex multiplication, which I'll summarize briefly.

DEFINITION 3.2. Let $E/\mathbf{C}$ be an elliptic curve. Then E has CM if the endomorphism ring $\text{End}(E)$ is larger than $\mathbf{Z}$.

LEMMA 3.3. Let $E/\mathbf{C}$ be an elliptic curve. Then $\text{End}(E) \neq \mathbf{Z}$ if and only if $\text{End}^0(E) = K/\mathbf{Q}$ is an imaginary quadratic field. In this case, $\text{End}(E) = \mathcal{O} \subset K$ is an order in K .

Proof. We may write $\text{End}(E) = \{\lambda \in \mathbf{C} : \lambda(\Lambda) \subseteq \Lambda\}$. Now assuming $\Lambda = \omega_1\mathbf{Z} \oplus \omega_2\mathbf{Z}$, to have $\lambda(\Lambda) \subset \Lambda$ means that $\lambda\omega_1 = a\omega_1 + b\omega_2$ and $\lambda\omega_2 = c\omega_1 + d\omega_2$ for $a, b, c, d \in \mathbf{Z}$. Then if $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, using the regular action we have $z = \omega_1/\omega_2$ which satisfies

$$z = \frac{\lambda\omega_1}{\lambda\omega_2} = \gamma \cdot z.$$

If $\lambda \notin \mathbf{Z}$, or E has CM, then b and c are nonzero, so $\mathbf{Q}(z)$ is an imaginary quadratic field by using the (quadratic) relation $z = \gamma \cdot z$. From the realizations of $\text{End}(E)$ and $\text{End}^0(E)$ in terms of Λ , we see $\text{End}^0(E) = \mathbf{Q}(z) = K$ is an imaginary quadratic field. It is clear $\text{End}(E)$ needs to be an order in K , since it is a ring which is a full and finitely generated $\mathbf{Z}$ -submodule of K . $\square$

Elliptic curves with CM can always be defined over $\bar{\mathbf{Q}}$, even K^{ab} .

In fact, the set of isomorphism classes over $\bar{\mathbf{Q}}$ of elliptic curves with CM by a particular order $\mathcal{O}$ is the same as the class group of that order.

THEOREM 3.4. Let $E/\mathbf{C}$ be an elliptic curve with CM by $\mathcal{O}_K$. Then $[\mathbf{Q}(j(E)) : \mathbf{Q}] \leq |\text{Cl}(\mathcal{O}_K)|$.

Proof. Since E has CM we can define it over $\bar{\mathbf{Q}}$. Write down the Weierstrass form of E , and apply any automorphism σ of $\bar{\mathbf{Q}}$ and note

$$j(E^\sigma) = j(E)^\sigma.$$

This also induces an equivalence $\text{End}(E) \simeq \text{End}(E^\sigma)$, so we again get an elliptic curve with CM. But there are at most $|\text{Cl}(\mathcal{O}_K)|$ such isomorphism classes, so $j(E)^\sigma$ can take on at most $|\text{Cl}(\mathcal{O}_K)|$ different values. This implies it is algebraic and also bounds the degree of $\mathbf{Q}(j(E))$. $\square$

In particular, the relevant fact for us is that we can always construct an elliptic curve with CM by $\mathcal{O}_K$, and in the situation that K has class number one then there is a unique such curve up to $\mathbf{Q}$ -isomorphism and $j(E) \in \mathbf{Q}$. In fact $j(E)$ must be an integer, as all j -invariants of CM elliptic curves are algebraic integers.

Idea. Reduce the class number one problem to finding points of a modular curve with integral j invariant.

We will need more conditions to actually make the number of such points finite.

Consider an imaginary quadratic field K of absolute discriminant D , and assume it has class number one. Then if $D > 28$, the prime 7 will actually remain prime. Otherwise, there is a prime (α) above 7 (it is principal by the class number assumption) and $\mathbf{N}(\alpha) = 7$. But when D is large, the integral basis for $\mathcal{O}_K$ forces α to be an integer, which is impossible given the norm condition.

Thus, we may assume 7 is inert.

LEMMA 3.5. If $D > 28$ and K has absolute discriminant D and class number one, there is an elliptic curve over $\mathbf{Q}$ whose base change has CM by $\mathcal{O}_K$ (which is unique up to $\bar{\mathbf{Q}}$ -isomorphism). The action of $\mathcal{O}_K$ on $E[7]$ gives it the structure of a 1-dimensional vector space over $\mathbf{F}_{49}$, and the $\mathbf{G}_{\mathbf{Q}}$ -action normalizes this nonsplit Cartan structure.

Proof. By the class number one condition, we know there must exist a unique isomorphism class over $\bar{\mathbf{Q}}$ of elliptic curves with CM by $\mathcal{O}_K$. Moreover, the j -invariant is an integer, so there is an elliptic curve over $\mathbf{Q}$ which lands in this isomorphism class over $\bar{\mathbf{Q}}$ (you can write a curve in Weierstrass form from the specified j -invariant, which completely classifies the isomorphism class over $\bar{\mathbf{Q}}$).

For the second statement, on $E[7]$ the $\mathcal{O}_K$ -action by endomorphisms becomes an action of $\mathcal{O}_K/7 \simeq \mathbf{F}_{49}$ (note that 7 in $\text{End}(E)$ actually corresponds to the multiplication by 7 endomorphism). But this is now a field, giving it a vector space structure. The full $\mathbf{G}_{\mathbf{Q}}$ -action need not commute with these endomorphisms; it normalizes the resulting nonsplit Cartan. $\square$

This extra structure on $E[7]$ is exactly what we need, and where the Klein quartic will come into play. The Klein quartic gives us explicit equations to describe $X(7)$. We will need a variant of $X(7)$ to capture this slightly different version of level structure.

DEFINITION 3.6. The 2-Sylow subgroup of $\mathrm{PSL}_2(\mathbf{F}_7)$ is isomorphic to the 8-element dihedral group D_8 , so we will denote it by this.

Let X denote the Klein quartic, and set $X_{\mathrm{ns}}(7) := X/D_8$.

THEOREM 3.7. The curve $X_{\mathrm{ns}}(7)$ is defined over $\mathbf{Q}$. Its non-cuspidal geometric points parameterize elliptic curves equipped with a non-split Cartan level structure, modulo the normalizer; equivalently, one may use a necklace of the cyclic order-7 subgroups of $E[7]$, up to reversal. Thus a rational point lying above an elliptic curve $E/\mathbf{Q}$ records a Galois-stable normalizer orbit, or equivalently, after choosing a basis of $E[7]$, the image of $G_{\mathbf{Q}}$ in $\mathrm{GL}_2(\mathbf{F}_7)$ is contained in the normalizer of a nonsplit Cartan. A non-split Cartan subgroup is a subgroup of the form

$$H = \left\{ \begin{pmatrix} a & \epsilon b \\ b & a \end{pmatrix} : a^2 - \epsilon b^2 \neq 0 \right\}$$

where $\epsilon \in \mathbf{F}_7^\times$ is a nonsquare. The normalizer of this subgroup maps to a dihedral group in $\mathrm{PGL}_2(\mathbf{F}_7)$, which is why the quotient by D_8 appears.

To set up the next part of the argument, we will consider the following diagram.

$$\begin{array}{ccc} X & \xrightarrow{\cong} & X(7) \\ \downarrow & & \downarrow \\ X/D_8 & \xrightarrow{\cong} & X_{\mathrm{ns}}(7) \\ \downarrow & & \downarrow \\ X/S_4 & & \\ \downarrow & & \downarrow \\ X/\mathrm{PSL}_2(\mathbf{F}_7) & \xrightarrow{\cong} & X(1) = \mathbf{P}^1 \end{array}$$

On the right hand side, the maps forget level structure with the map to $X(1)$ being the j -invariant. The idea is that we will produce points of $X_{\mathrm{ns}}(7)$ with integral j -invariant, and then translate across this diagram to turn this into a diophantine equation with finitely many solutions.

The easiest step is to write down the map $j : X \rightarrow X(1) \simeq \mathbf{P}^1$ in terms of the coordinates x, y, z as a rational function of degree 168 (there is a simpler way to express it in terms of certain invariant polynomials coming from the representation theory of $\mathrm{PSL}_2(\mathbf{F}_7)$).

Our goal is to give an explicit rational parameter ϕ for X/D_8 along with the j map, so that we can write j in terms of this parameter and ask for rational solutions so $j \in \mathbf{Z}$.

This is done by first describing the genus zero curve X/S_4 as rationally parameterized by an explicit coordinate ψ on X . The j -function can also be explicitly given, describing the map $X/S_4 \rightarrow X/\mathrm{PSL}_2(\mathbf{F}_7) \simeq X(1) = \mathbf{P}^1$. To do this one writes down the j function on $X/\mathrm{PSL}_2(\mathbf{F}_7)$ as a degree 168 rational function, and then write j as a rational function of the coordinate ψ on X/S_4 .

Then, we can describe X/D_8 as a degree 3 cover of X/S_4 (it will in fact be genus 0 again). This gives it a coordinate ϕ where we can write $\psi \in \mathbf{Q}(\phi)$. In total, we obtain in all its horrible glory that

$$j = 12^3 + 56^2 \frac{(\phi - 3)(2\phi^4 - 14\phi^3 + 21\phi^2 + 28\phi + 7)P^2(\phi)}{(\phi^3 - 7\phi^2 + 7\phi + 7)^7}$$

where $P(\phi) = (\phi^4 - 14\phi^2 + 56\phi + 21)(\phi^4 - 7\phi^3 + 14\phi^2 - 7\phi + 7)$.

THEOREM 3.8. The only imaginary quadratic fields with absolute discriminant D whose rings of integers are PIDs occur when

$$D = 3, 4, 7, 8, 11, 19, 43, 67, 163.$$

Proof. As we have seen, once $D > 28$ we may assume 7 is inert and then produce an elliptic curve E over $\mathbf{Q}$ whose base change has CM by $\mathcal{O}_K$ and whose j -invariant is an integer. Moreover, we found that the Galois action on $E[7]$ normalizes the nonsplit Cartan coming from the $\mathbf{F}_{49}$ -vector space structure. This means that E gives a rational point of $X_{\mathrm{ns}}(7)$ (in particular a $\mathbf{Z}[1/7]$ -point which does not give cusps for $p \neq 7$) such that the j -invariant is an integer.

Now we can use the massive formula from earlier, together with the finite computation of the integral points on the level-7 nonsplit-Cartan curve, to solve the problem. Put $\phi = \frac{m}{n}$ with $\gcd$ one. Writing

$$j = \frac{A(m, n)}{B(m, n)}$$

we find that $\gcd(A(m, n), B(m, n)) | 56^7$ given $(m, n) = 1$. In particular,

$$(m^3 - 7m^2n + 7mn^2 + 7n^3) | 56.$$

This reduces the remaining step to a finite computation. Essentially, m/n needs to be a good rational approximation on the order of n^3 of a root α of $\phi^3 - 7\phi^2 + 7\phi + 7$, as

$m^3 - 7m^2n + 7mn^2 + 7n^3$ is forced to be small. Any bound

$$|\alpha - m/n| > C_\alpha n^{-C'_\alpha}$$

where $C'_\alpha < 3$ suffices, and many exist (in this case 0.099 and $7/3$). This bounds n , after which the remaining coprime pairs (m, n) can be checked directly; equivalently, the same finite list can be translated into Nagell's exceptional units in $\mathbf{Q}(\cos(2\pi/7))$. The resulting integral j -values give only the CM discriminants already displayed above, and combining this with the small cases $D \leq 28$ gives exactly the stated list. $\square$